

$$P_{yy}^{\infty(j)}(f) = \frac{1}{MU} \left| \sum_{m=0}^{M-1} y_j(m) y^k w(m) e^{-j2\pi f n} \right|^2 \quad (6)$$

In this case, the window function's normalizing factor of the power is represented by U , which is signified in equation (7):

$$U = \frac{1}{M} \sum_m^{M-1} W^2(m), \quad (7)$$

Step 3: In this step, EEG segmentation is performed based on trial and subject resulting in 72 EEG images for each case which is described in Figure 4, respectively.

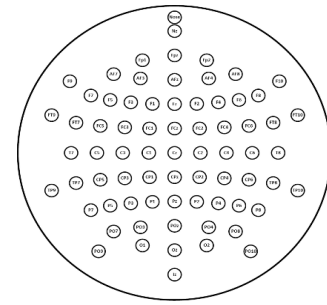


Fig 4: Layout of 10-20 system 72 EEG channel electrode placement system [25].

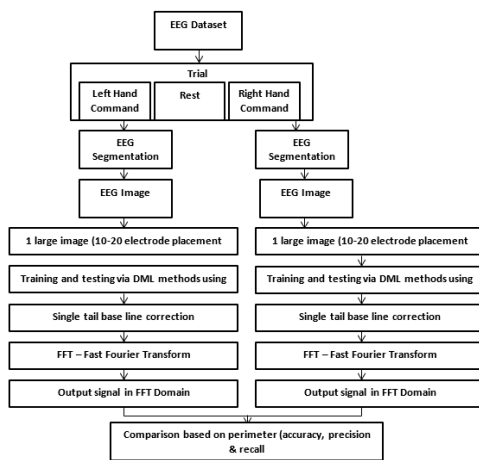


Fig 2: Research Methodology

Step 1: The EEG dataset which is used consists of 72 EEG channels according to a ten is to twenty electrode placement system which is recorded from twenty-two people considered as test subjects.

Step 2: Everyone trailed for the right hand or left hand to move on the right side of the display or left side of the display. The trials with no target are labeled as rest. In Figure 3 FFT power spectrogram of a single trail for right-hand movement, imagination has been presented for subject 19.

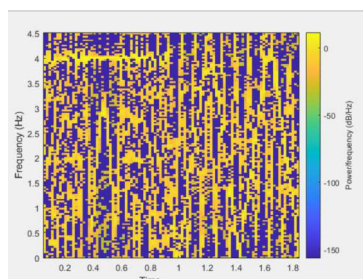


Fig 3: FFT power spectrogram of right-hand movement of subject 19

Fig 5: The combined image of 72 channels of a single trail.

Step 5: After that, training and testing are performed via the DML method using the nearest neighbor kernel method over these two large images.

Step 6: Then, single trail baseline correction is applied to improve the SNR.

Step 7: After that, time domain to frequency signal conversion is accomplished by the exploitation of the FFT.

